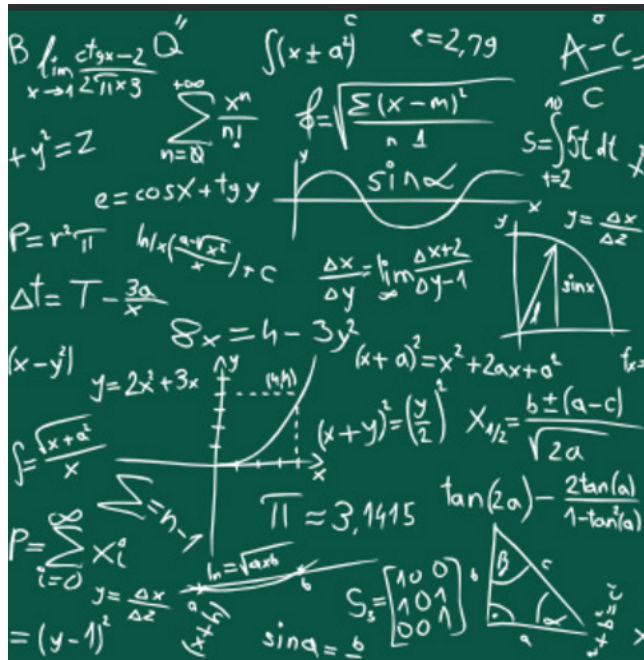


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Domains of functions



- **Author:** Fabio Furini.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

Exercise 1

Determine the domain of the function:

$$f(x) = \frac{\log_2(3+x)}{\sqrt[3]{x+2}}$$

it is determined by studying the numerator and the denominator of the function

- The numerator:

The function $\log_2 t$ is defined for $t > 0$, hence the function $\log_2(3+x)$ is defined for $3+x > 0$, i.e., $x > -3$.

- The denominator:

The function $\sqrt[3]{t}$ is always defined; however, since it is in the denominator, it must not vanish. Hence we impose $\sqrt[3]{x+2} \neq 0$, i.e., $x \neq -2$.

Hence the domain is

$$(-3, -2) \cup (-2, +\infty)$$

Exercise 2

Determine the domain of the function:

$$f(x) = \sqrt{2^x - 8} \cdot \log|x - 4|$$

Exercise 3

Determine the domain of the function:

$$f(x) = \tan(\log x)$$

Exercise 4

Determine the domain of the function:

$$f(x) = \log(\tan x)$$